$$2PbS + 3O_2 \xrightarrow{\Delta} 2PbO + 2SO_2\uparrow$$

$$2ZnS + 3O_2 \xrightarrow{\Delta} 2ZnO + 2SO_2\uparrow$$

$$2Cu_2S + 3O_2 \xrightarrow{\Delta} 2Cu_2O + 2SO_2\uparrow$$

Roasting also removes impurities such as arsenic, sulphur, phosphorous by converting them into their volatile oxides.

For example

$$4As + 3O_2 \longrightarrow 2As_2O_3\uparrow$$

$$S_8 + 8O_2 \longrightarrow 8SO_2\uparrow$$

$$P_4 + 5O_2 \longrightarrow P_4O_{10}\uparrow$$



The sulphur dioxide produced during roasting process is harmful to the environment. In modern metallurgical factories, this by product is trapped and converted into sulphuric acid to avoid air pollution.

Calcination

Calcination is the process in which the concentrated ore is strongly heated in the absence of air. During this process, the water of crystallisation present in the hydrated oxide escapes as moisture. Any organic matter (if present) also get expelled leaving behind a porous ore. This method can also be carried out with a limited supply of air.

For examples,

During calcination of carbonate ore, carbon dioxide is expelled

$$PbCO_3 \xrightarrow{\Delta} PbO + CO_2\uparrow$$

$$CaCO_3 \longrightarrow CaO + CO_2\uparrow$$

$$ZnCO_3 \xrightarrow{\Delta} ZnO + CO_2\uparrow$$

$$MgCO_3.CaCO_3 \xrightarrow{\Delta} MgO + CaO + 2CO_2\uparrow$$

During calcination of hydrated ore, the water of hydration is expelled as vapour

XII U1 Metallurgy - Jerald.indd 7 16-12-2022 18:52:59